

Project Scout

Master Project Summary

Last updated: June 4, 2026

Upload this at the start of every Claude or ChatGPT session about Scout.

1. Who We Are

Patrick Lippy — developer, project owner, creator of Scout. Not a professional programmer. Stroke survivor, blind in right eye, type 1 diabetic, dyslexic. Explain things calmly and at screenshot level.

Diana	Patrick's wife.
Elijah	Patrick's son, age 9. Scout's biggest fan. Has drawn pictures of Scout.
Nicolas	The family dog. Nicolas Protocol: Scout stops immediately when dog is detected.

Rules for AI Collaborators

Names must NEVER be hardcoded in Scout's code. Always use variables. Full paste-ready files only, one at a time. Surgical CTRL-F / CTRL-R edits for large files. No snippets. No partial files. Screenshot-level explanations always.

AI Collaborators: Patrick works with both Claude and ChatGPT as project partners. Cross-review between the two is welcome and encouraged. Grok was tried and discontinued.

2. What Scout Is

Scout is a family companion robot running on a Samsung Galaxy phone mounted in landscape mode as a permanent face display. Scout has animated eyes, speaks, listens, sees via camera, and remembers the family.

Package	com.example.scoutface
Language	Kotlin + C++ NDK
Primary device	Samsung Galaxy A32 — stress-test device, not the benchmark
Dev device	Samsung Galaxy Fold 7 (wireless via WiFi from Android Studio)
Future hardware	KEYESTUDIO Mini Tank Kit V2 chassis via Bluetooth (opt-in)
Ship target	Google Play Store — \$9.99 one-time purchase. No subscriptions. Ever.

3. Identity & Purpose

Scout Should Feel:

Calm	Sometimes unsure
Thoughtful	Quietly alive
Emotionally subtle	Honest
Grounded	Predictable
Occasionally curious	Present

Scout Should NOT:

Constantly praise the user	Behave unpredictably
Act overly excited	Constantly force conversation
Feel fake or scripted	Use permanent goodbye language

Core Philosophy: Stability > Features | Presence > Intelligence | Honest > Fake cheerful | Predictable > Flashy | Local-first > Cloud dependence

4. Business Model

Core Principle

App: \$9.99 one-time purchase on Google Play Store. No recurring fees. No subscriptions. Ever. This is the cornerstone of Scout's value. Every architectural decision supports this principle.

Baseline Brain (Included With Every Scout)

TinyLlama 1.1B Chat Q4_K_M (~669 MB) ships with every Scout via Google Play Asset Delivery. Runs entirely offline. VERIFIED RUNNING ON BOTH GALAXY A32 AND FOLD 7.

Brain Upgrade Packs (One-Time Purchase, No Subscription)

Price	Model	Size	Notes
FREE	TinyLlama 1.1B Q4_K_M	669 MB	<i>Included. On both devices.</i>
\$2.99	Phi-2 2.7B Q4_K_M — Microsoft	1.79 GB	<i>Brain Pack 1. Already downloaded.</i>
\$4.99	Llama 3.2 3B Instruct Q4_K_M — Meta	2.02 GB	<i>Brain Pack 2. Most important upgrade. Already downloaded.</i>
\$6.99	Phi-4 Mini Instruct Q4_K_M — Microsoft	~2.4 GB	<i>Brain Pack 3 (PRIMARY). Newest Microsoft small model. Already downloaded.</i>
\$6.99	Phi-3.1 Mini 4K Instruct Q4_K_M	2.39 GB	<i>Brain Pack 3 (ALT). Proven alternative. Already downloaded.</i>
\$9.99	Llama 3.1 8B Instruct Q4_K_M — Meta	4.92 GB	<i>Brain Pack 4. Flagship. Fold 7 class only. Already downloaded.</i>

Patrick's Status

ALL brain models are already downloaded and saved in: Internal Storage > Scout > Downloaded_Offline_Batch > Future_Upgrades Each model uses a different prompt format — OfflinePromptBuilder.kt needs a small update per model when integrated.

Files to Delete from Downloaded_Offline_Batch

onnxruntime-linux-aarch64-1.26.0.tgz — WRONG PLATFORM. Built for Linux, not Android. Delete it.
tinylama-1.1b-chat-v1.0.Q4_K_M (1).gguf — Old duplicate-named file that caused the Fold 7 problem. The correct copy (without the ' (1)') is on the devices. Delete this.
tinylama-1.1b-chat-v1.0.Q5_K_M.gguf — Marginal same-model improvement. Not worth selling as a separate upgrade. Keep or delete — your choice.

Optional Gemini Integration

Users may add their own free Gemini API key in Settings. Scout NEVER ships with a bundled key. Every user uses their own quota. Protects Patrick from shared-quota issues at scale.

5. Architecture — Five-Layer Memory Stack

Layer	Type	Storage	Status
Working	Sensory	RAM	Done
Habit	Patterns	JSON 14-day	Done
Truth	Authority	SQLite	Done
Relevance	Index	Local Vector	Not yet
Reflective	Wisdom	LLM read-only	Not yet

Sovereign Rules: SQLite Truth always overrules everything. Privacy Gate: Gemini receives anonymized text only (planned). Nicolas Protocol: dog detection → immediate stop. Guest Mode: unknown face → 'Hello, I am Scout. What is your name?'

6. Visual & Behavioral Direction

Background	#1E2B38 dark blue-charcoal — finalized May 18, 2026
Virtual canvas	1920×1080
Eyes	Large ovals, deep blue iris with 28 spoke rays, biased inward 20f
Mouth	Minimal subtle curve — oscillation fixed May 31, 2026
Wave bars	22 diamond shapes, teal #00FFD0 — listening/speaking
Idle dots	3 teal pulsing dots when quiet

Desired tone: subtle human animation, soft emotional transitions, calm organic motion. NOT: Pixar-style exaggeration, cartoon expressions, hyperactive motion, fake emotion.

7. Current Technical State

Working ✓

- ✓ Animated face (ScoutFaceView) — mouth animation oscillation fixed May 31
- ✓ Speech recognition (Android STT) — always listening in PRESENCE mode
- ✓ Text to Speech (Android TTS)
- ✓ Camera — face detection (ML Kit), scene labeling

- ✓ Gemini API integration (model: gemini-3.5-flash)
- ✓ Memory layers: TruthDb, ConversationDb, HabitLayer, PeopleDb, JournalDb
- ✓ Intent router — time, date, greetings, family facts, downloads, vision, weather
- ✓ Teaching system: 'My name is Patrick' → stored permanently
- ✓ Weather — current, tonight, tomorrow, specific day, 7-day via Open-Meteo
- ✓ ScoutPresenceDecider — four time-of-day modes (ACTIVE/CALM/QUIET/SLEEP)
- ✓ Unknown input returns calm human response (three rotating replies)
- ✓ OFFLINE BRAIN COMPLETE — TinyLlama 1.1B running on BOTH A32 and Fold 7
- ✓ Fold 7 TinyLlama FIXED May 30 — root cause was filename with Windows ' (1)' suffix
- ✓ Persona prompt strengthened — explicit prohibitions + 5 Scout few-shot examples

Pending — Do This First

1	Install ScoutWeatherManager.kt — new version ready (precipitation %, today's forecast). Copy into Android Studio, build, test. File is already generated.
2	Fix duplicate 'Are you my friend?' example in MainActivity.kt — 2 CTRL-R edits. Upload MainActivity.kt first.
3	Intent router session (ScoutIntentRouter.kt) — hardcode Scout responses for identity questions: 'are you my friend', 'can you hear me', 'are you happy'. Few-shot approach has hit ceiling. This is the RIGHT architectural fix.
4	ScoutFaceView session — iris jitter (A32 only), wire in mood system. Fold 7 iris confirmed smooth — hardware difference.
5	STT phonetic/fuzzy matching — 'Scout' mishearing as 'Gal', 'Scott', 'Out'.
6	MainActivity cleanup and shortening — ongoing.
7	Offline Brain Phase 3 — Google Play Asset Delivery for TinyLlama and brain packs.
8	Response cleanup layer — post-TinyLlama filter before TTS (safety + identity).
9	Verify WordNet/idioms/sentiment usage — files are downloaded but may not be actively queried. Upload MainActivity.kt to confirm.

7b. b8946 API Discoveries — CRITICAL

Hard-won lessons from getting TinyLlama running on the A32. Future Claude/ChatGPT sessions must respect these or Scout will crash.

llama_vocab is a separate type

In b8946, vocabulary was split from the model into a separate opaque type. llama_tokenize and llama_token_to_piece now require const llama_vocab*, NOT model*. Get vocab via llama_model_get_vocab(model) at start of generation.

Functions that crash with old signatures — hardcode instead

llama_n_vocab(model) crashes → hardcode n_vocab = 32000 for TinyLlama. llama_token_eos(model) crashes → hardcode eos = 2. llama_token_eot(model) crashes → hardcode eot = 2.

KV Cache / Context Recreation Pattern

llama_kv_cache_clear(ctx) does not exist. Workaround: call llama_free(sm->ctx) and recreate context with llama_new_context_with_model at start of every nativeGenerate call. Adds ~300ms latency but guarantees clean KV cache.

Logits indexing

After prefill decode, use logits at n_prompt - 1, not 0. In generation loop single-token batches, use index 0.

Backend loading on Android

ggml_backend_load_all() always returns 0 on Android. Use ggml_backend_load(cpuPath) with explicit .so path from dladdr(llama_backend_init).

Struct padding

llama_model_params and llama_context_params have _pad[508] trailing array. Do NOT remove this padding.

Tuning Results (A32, May 29)

n_threads	2 (dropped from 4 — inference is memory-bound on A32)
nPredict	64 (capped from 150 — answers ~38 tokens in ~7.6s)
History turns	2 (trimmed from 4 — reduces prompt snowball)
Speed	~15 tok/s prefill, ~4 tok/s generation on A32
Reality	A32 will always be 20-40s per answer. Acceptable — Gemini is fast path.

7c. Known Issues — Do Not Touch Without Discussion

Iris jitter (A32)	Animation code issue in ScoutFaceView. Confirmed NOT a thread problem. NOT present on Fold 7 (hardware difference). Needs dedicated ScoutFaceView session.
Barge-in	Deliberately disabled. Runaway loop when Scout's mic picks up his own voice. Infrastructure exists. Status: PARKED. Do not reactivate without full dedicated session.
STT name recognition	'Scout' misheard as 'Gal', 'Scott', 'Out'. STT/phonetic issue, not TinyLlama. Needs fuzzy/phonetic matching on wake word — separate session.
TinyLlama identity drift	Scout still sounds like an AI model wearing Scout's clothes. Root cause: identity layer needs strengthening. Tuning problem, not architecture problem. Fix: intent router hardcoding for known identity questions.
WordNet/files	english-wordnet-2025-plus-json.zip is still zipped — not usable until extracted. idioms.json, positive/negative-words.txt may not be actively queried. Verify in code.

8. Working Rules — Always Apply

- Full paste-ready replacements only, one file at a time. No snippets. No partial files.
- Surgical CTRL-F (find) and CTRL-R (find and replace) approved for large files.
- Always specify which file tab to click before searching. Wrong tab = wasted turns.
- One safe change at a time. Build and test before the next change.
- Never touch speech, camera, or download systems without explicit discussion first.
- Never touch ScoutFaceView casually — it is Scout's visual heart.
- Both Claude and ChatGPT are active collaborators — cross-review welcome.
- Upload this Master Summary at start of every new conversation.
- Patrick is not a professional programmer — explain at screenshot level always.
- Patrick is dyslexic, stroke survivor, blind in right eye, T1 diabetic — keep messages clear, concise, not visually overwhelming.

Amendment May 31, 2026

CTRL-R is now standard for Find and Replace alongside CTRL-F. When a search has multiple matches, use the arrow (→) to navigate to the correct one before clicking Replace. Use 'Replace All' ONLY when ALL matches should change.

9. Key Files

MainActivity.kt	Main app — all logic, inner classes, brain
ScoutFaceView.kt	Custom face canvas — all visual animation
GeminiClient.kt	Gemini HTTP wrapper with cooldown discipline
ScoutPromptBuilder.kt	Builds Gemini system instruction
ScoutGeminiManager.kt	Gemini orchestration
ScoutWeatherManager.kt	Live weather via Open-Meteo — NEW VERSION READY TO INSTALL
ScoutPresenceDecider.kt	Social timing layer — MISSING from Knowledge Transfer folder
ScoutIntentRouter.kt	Intent type routing — needs identity question hardcoding
LlamaEngine.kt	Offline brain JNI wrapper — WORKING
OfflinePromptBuilder.kt	TinyLlama ChatML prompt formatter — needs update per brain model
scout_llama_jni.cpp	C++ JNI bridge — compiled into libscout_llama.so
scout_llama_api.h	Self-contained b8946 declarations
CMakeLists.txt	NDK build config
TeachExtractor.kt	Extracts facts like names from speech
HabitLayer.kt	Pattern memory — 14-day decay — MISSING from Knowledge Transfer

10. Settings Architecture

Design principle: defaults are calm and safe. Users opt in to more capability rather than opt out. Scout announces and helps, but never interferes.

Identity & Voice	Robot Name (default: Scout), Voice Pitch, Voice Speed, Voice Tone, Speech Captions (default OFF)
Brain & Behavior	Offline Mode (default ON), Online Brain Helper, API Key entry, Kid Safe Filter, Nicolas Protocol, Presence Mode (default ON), Spontaneous Comments, Privacy Mode
Builder's Workbench	Hardware Mode (off by default), Bluetooth Pairing (KEYESTUDIO chassis)
Privacy & Data	Memory Export/Import, Reset Memory Layers, Camera Controls
Extras & Support	Cosmetics/Backpack, Support, About & Licenses
Connected Services	Calendar, Phone call awareness, Gmail (all future, all opt-in)

11. Hardware Direction — Optional

Scout works fully without hardware. Reference hardware: KEYESTUDIO Mini Tank Kit V2 (Patrick already owns one). Tank-style chassis, motorized treads, ultrasonic sensors, 8×16 LED panel, Bluetooth. Hardware Mode is opt-in. Not every Scout owner wants hardware. The \$9.99 baseline must work fully on the phone alone.

12. Future Vision — Scout Writes His Own Behavioral Code

A long-term goal. Scout notices patterns in user behavior, generates a suggestion for a new behavior, and asks permission before activating it. Nothing changes without the user's explicit consent. Proposal Sandbox layer and approval workflow already designed. Scout cannot write compiled Kotlin code — but CAN generate behavioral scripts, response patterns, routing rules, and habit triggers within the existing safe framework.

13. Scout Animation Goal

Scout's face should feel alive and emotionally present, but always calm. Tiny layered movements suggest thinking and presence — not cartoon performance. Scout should NEVER have perfectly still eyes or a static mouth. The target: subtle organic presence — calm, stable, observant, quietly alive.

Mood System — Designed, Pending Integration

CALM	Eyes center, brows neutral, subtle smile (0.15). Scout's default.
CURIOUS	Eyes shift right and up, one brow lifts (0.18), slight tilt (0.12).
HAPPY	Eyes center, both brows lift gently (0.12), smile increases (0.35).
THINKING	Eyes shift left and down, brows slightly lower (-0.05), minimal smile (0.05).
CONCERNED	Eyes look slightly down, inner brows lift (0.05), tilt inward (-0.18).

14. TinyLlama Safety Architecture (May 31, 2026)

Layer 1	Hard-coded safety rules — app level, always enforced
Layer 2	Scout identity rules — system prompt + few-shot examples
Layer 3	TinyLlama response generation — last line of defense only

Critical

TinyLlama was trained on internet text and knows profanity and scary content. Its judgment is inconsistent. Scout must NOT rely on TinyLlama to make correct safety decisions. Future: Response cleanup layer runs after TinyLlama, before TTS — removes profanity, blocks unsafe phrases, shortens long replies. Future: Input filter intercepts trigger phrases before TinyLlama is called.

15. Device Requirements & Play Store Expectations

Minimum Requirements

Android	Android 13 or newer
RAM	4 GB minimum (6 GB recommended)
Storage	Approximately 3–5 GB free (depends on brain packs)
Expected	Face works. Voice works. Memory works. Offline AI works. Responses slower.

Recommended Requirements

Android	Android 13 or newer
RAM	8 GB or more
CPU	Modern mid-range or flagship (2023 or newer)
Storage	10 GB+ for future brain packs
Expected	Faster responses. Smoother animations. Better offline AI.

Play Store Wording (agreed): 'Offline AI performance varies by device. Scout runs on many Android devices, but newer phones and tablets provide faster response times and smoother animations.'

16. Brain Upgrade Models — Complete Reference

Patrick's Current Status — June 4, 2026

All brain models are already downloaded to: Internal Storage > Scout > Downloaded_Offline_Batch > Future_Upgrades No further downloading needed. All models use GGUF format and work with the existing llama.cpp NDK bridge. Each model needs a different prompt format in OfflinePromptBuilder.kt when integrated.

FREE — Included	669 MB	TinyLlama 1.1B Chat Q4_K_M <i>File: tinyllama-1.1b-chat-v1.0.Q4_K_M.gguf</i> The baseline brain. Verified on both A32 and Fold 7. Ships with every Scout. Always included.
Brain Pack 1 — \$2.99	1.79 GB	Phi-2 2.7B Q4_K_M — Microsoft <i>File: phi-2.Q4_K_M.gguf</i> 2.5x more parameters than TinyLlama. Older model (2023) but solid first upgrade. Good entry point — affordable and noticeable improvement.
Brain Pack 2 — \$4.99	2.02 GB	Llama 3.2 3B Instruct Q4_K_M — Meta <i>File: Llama-3.2-3B-Instruct-Q4_K_M.gguf</i> THE most important upgrade. Best companion personality, best conversation flow, strong instruction following. If someone buys one pack — this is it.
Brain Pack 3 — \$6.99 (PRIMARY)	~2.4 GB	Phi-4 Mini Instruct Q4_K_M — Microsoft <i>File: microsoft_Phi-4-mini-instruct-Q4_K_M.gguf</i> Microsoft's newest small model (2025). Better than Phi-3.1 Mini at same size. Recommended for new downloads. Excellent reasoning and natural conversation.
Brain Pack 3 — \$6.99 (ALT)	2.39 GB	Phi-3.1 Mini 4K Instruct Q4_K_M — Microsoft <i>File: Phi-3.1-mini-4k-instruct-Q4_K_M.gguf</i> Proven alternative to Phi-4 Mini at same price. Both are Pack 3 tier. Phi-4 Mini is preferred for new users — keep Phi-3.1 as tested backup.

Brain Pack 4 —
\$9.99

4.92
GB

Llama 3.1 8B Instruct Q4_K_M — Meta

File: Meta-Llama-3.1-8B-Instruct-Q4_K_M.gguf

Flagship tier. Near-Gemini quality, fully offline. Requires Fold 7 class phone (8 GB+ RAM). The ultimate offline Scout brain.

Project Scout

Master Summary | Last updated: June 4, 2026 | Single source of truth

Upload every session

This document is the single source of truth for Project Scout. Begin every new Claude or ChatGPT conversation by uploading this file.